PACLAND · Belief, in plain words

Verification says what is proven. This says what to believe about the rest, and what to check next.

1 · The question

Verification has a clear job: proven or refuted, with a machine-checked reason. But most interesting claims are still open. What should you believe about them, and which should you check next?

2 · What we found

- “How sure are you” has to be a number that behaves like a probability — otherwise a set of bets can lose you money no matter what. A belief is a number from 0 to 1.
- A test moves the number a fixed way (Bayes' rule): new belief equals old belief times how much more likely the result is if you are right. A weak test moves it a little; a strong one a lot.
- You can rank “what to check next” first. Each claim scores how much the next test would move its belief, divided by cost. The top of the list is the next job.
- “Confidence” is itself checkable: say “70% sure” a hundred times and you should be right about 70. Scored that way, our own early confidence was overconfident — as experts usually are. A ledger records it.
- Some questions never settle. When data cannot separate two answers, the honest output is the whole set, weighted by a prior, never one fake answer.

3 · The best we have right now

A running belief layer with four parts, plus a detector that knows which of two situations a claim is in: “converge” (shared evidence pulls everyone to one answer) or “plural” (it never will, and the report says so). And a guarantee: a belief carries a probably-approximately-correct bound, so “I believe 70%” comes with a stated margin.

4 · How we refine it

- Score more beliefs against more outcomes, so the calibration number outgrows “n is small”.
- State every prior as a prior, never as a finding.
- Run the detector first: a plural claim reports the weighted set, not a fake point.

5 · What this means for the world

A decision engine, not just a lie detector. Verification says what is proven; belief says what is worth believing and testing next — so a researcher, a regulator, or an AI spends its next hour where it learns the most.

6 · What this cannot do

It cannot make an unproven claim proven, or conjure a value that was never in the data. It is honest about its limits — which is why the number can be trusted.